THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE CHAPTER TEST SET B

Chapter 12: **Magnetic Effects of Electric Current** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A compass needle is a: 1

(a) coil **(b)** piece of iron **(c)** conductor **(d)** small bar magnet

2. Inside a bar magnet, field lines go from: 1

(a) they do not exist **(b)** north to south
(c) south to north **(d)** east to west

3. The field produced by a solenoid resembles that of a: 1

(a) straight wire **(b)** bar magnet
(c) circular loop only **(d)** horse-shoe magnet

4. A student states: “The magnetic field at the centre of a circular loop increases with the number of turns.” The correct response is: 1

(a) Correct **(b)** Incorrect
(c) Only for straight wires **(d)** Cannot be determined

5. For an electron beam moving in a magnetic field, Student A says it experiences a force; Student B says it does not. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

6. The potential difference between live and neutral wire in India is: 1

(a) 440 V **(b)** 110 V **(c)** 220 V **(d)** 12 V

7. The live wire usually has insulation of colour: 1

(a) black **(b)** yellow **(c)** green **(d)** red

8. Which of these is used to make a strong electromagnet? 1

(a) copper core **(b)** steel core **(c)** soft iron core **(d)** wooden core

9. In Fleming's left-hand rule, the middle finger indicates the direction of: 1

(a) force **(b)** current **(c)** motion **(d)** magnetic field

10. The SI unit of magnetic field strength is: 1

(a) gauss **(b)** tesla **(c)** weber **(d)** ampere

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Why does a compass needle get deflected near a bar magnet? 2

12. Name two safety devices used in domestic circuits. 2

13. What is the function of the earth wire? 2

14. A student says the field at the centre of a circular loop is zero. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Draw (describe) the pattern of field lines around a bar magnet. List two properties of field lines. 3

16. State the factors on which the strength of the magnetic field of a solenoid depends. 3

17. A current-carrying conductor is placed in a magnetic field. When is the force on it 3
(i) maximum
(ii) zero? Explain the force direction using a rule.

OR

Explain why a fuse is connected in series with the live wire. What is the fuse made of? 3

18. What precautions should be taken to avoid overloading and short-circuiting? 3

OR

Compare the field due to a straight conductor and a solenoid. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Explain the magnetic field due to a solenoid with a diagram. 5
(ii) How can a solenoid be used to magnetise a steel rod?
(iii) Difference between electromagnet and permanent magnet.

OR

(i) State Fleming's left-hand rule with a diagram. 5
(ii) A proton moves in a magnetic field directed into the page; it moves to the right. Find the direction of force.
(iii) Name two devices using this effect.

20. **Case study:** A house's MCB tripped when a geyser, heater, AC and iron were switched on together. 5
(i) Name the problem.
(ii) Why did the MCB trip?
(iii) How is MCB better than a fuse?
(iv) Suggest a safe practice.
(v) What would happen without an MCB?

OR

(i) Describe an activity to show the magnetic field around a straight current-carrying conductor. 5
(ii) What happens to the pattern when the current is increased?

END OF QUESTION PAPER — Total: 40 Marks